Name: _____

Class: _____

Date: _____



Mr. Rawson • GCSE Physics

Kinematics 7 Supplement: the four forms of $v^2 - u^2 = 2\,a\,s$ — ANSWERS

1. Order of operations — BIDMAS

Always remember BIDMAS!

	Stands for	What to do	Example
B	Brackets	calculate inside brackets first	$(3+2) \times 4 = 5 \times 4 = 20$
I	Indices (powers)	then do exponents	$3 + 4^2 = 3 + 16 = 19$
D M	Division, Multiplication	then $\div$ or $\times$, left to right	$12 \div 3 \times 2 = 4 \times 2 = 8$
A S	Addition, Subtraction	then + or - , left to right	$10 - 3 + 2 = 7 + 2 = 9$

More Tips:

- **FRACTIONS -** consider these like Brackets. Do the top calculations, then the bottom calculations, then divide the whole thing.
- **SQUARE ROOT -** A square root sign works like brackets. Work out everything under the $\sqrt{\ }$, then calculate the $\sqrt{\ }$ after.

2. SUVAT — what each letter means

Symbol	Name	Unit	Unit in words
s	displacement (distance)	m	metres
u	initial velocity	m/s	metres per second
v	final velocity	m/s	metres per second
a	acceleration	m/s^2	metres per second squared
t	time	s	seconds

A. Find the distance, s

Worked example A — find the distance

A bus speeds up from $4.0\,\text{m/s}$ to $12\,\text{m/s}$ with an acceleration of $2.0\,\text{m/s}^2$. Calculate the distance it travels while speeding up.

What the question tells you

Initial velocity	$=u=$	$4.0\,\text{m/s}$	
Final velocity	$=v=$	$12\,\text{m/s}$	
Acceleration	$=a=$	$2.0\,\text{m/s}^2$	
Distance	$=s=$	**?**	*the one the question asks for*
Time	$=t=$		*not in the question: circle t in the Givens and cross it out*

1. Givens

$s =$ **?**
$u = 4.0$ m/s
$v = 12$ m/s
$a = 2.0$ m/s^2
~~t~~ =

2. Equation $v^2 - u^2 = 2 \quad a \quad s$

3. Substitute $12^2 - 4.0^2 = 2 \times 2.0 \times s$

4. Rearrange $s = \dfrac{12^2 - 4.0^2}{2 \times 2.0}$ *form A*

5. Calculate

$$v^2 = 12^2 = 144$$
$$u^2 = 4.0^2 = 16$$
$$v^2 - u^2 = 144 - 16 = 128$$
$$2 \times a = 2 \times 2.0 = 4.0$$
$$s = 128 \div 4.0 = \mathbf{32\ m}$$

Check: $v^2 - u^2 = 128$ and $2\,a\,s = 2 \times 2.0 \times 32 = 128$. Same number.

Now try. Fill in the Givens, circle and cross out t, then work down the ladder, one line at a time.

A1. A car speeds up from $10\,\text{m/s}$ to $20\,\text{m/s}$ with an acceleration of $3.0\,\text{m/s}^2$. Calculate the distance.

Givens

$s =$?
$u = 10$ m/s
$v = 20$ m/s
$a = 3.0$ m/s^2
~~t~~ =

Form: $s = \dfrac{v^2 - u^2}{2\,a}$

$v^2 =$ $20^2 = 400$

$u^2=$ $10^2 = 100$

$v^2 - u^2 =$ $400 - 100 = 300$

$2 \times a =$ $2 \times 3.0 = 6.0$

$s =$ $300 \div 6.0 =$ ***50 m***

Check: $v^2 - u^2 =$ *300* $\quad 2\,a\,s =$ *300*

B. Find the acceleration, a

Worked example B — find the acceleration

A sprinter speeds up from 3.0 m/s to 9.0 m/s over a distance of 24 m. Calculate the sprinter's acceleration.

1. Givens

$s = 24$ m
$u = 3.0$ m/s
$v = 9.0$ m/s
$a =$ **?**
$\not{x} =$

2. Equation $\quad v^2 - u^2 = 2 \quad a \quad s$

3. Substitute $\quad 9.0^2 - 3.0^2 = 2 \times a \times 24$

4. Rearrange $\quad a = \dfrac{9.0^2 - 3.0^2}{2 \times 24}$ *form B*

5. Calculate

$$v^2 = 9.0^2 = 81$$
$$u^2 = 3.0^2 = 9.0$$
$$v^2 - u^2 = 81 - 9.0 = 72$$
$$2 \times s = 2 \times 24 = 48$$
$$a = 72 \div 48 = \mathbf{1.5\ m/s^2}$$

Check: $v^2 - u^2 = 72$ and $2\,a\,s = 2 \times 1.5 \times 24 = 72$. Same number.

Now try. In form B the bottom line is $2 \times s$.

B1. A bike slows from 8.0 m/s to a stop over 16 m. Calculate the acceleration. Slowing down gives a negative acceleration, so your answer has a minus sign.

Givens

$s = 16$ m
$u = 8.0$ m/s
$v = 0$
$a =$?
$\not{x} =$

Form: $a = \dfrac{v^2 - u^2}{2\,s}$

$v^2 =$ $0^2 = 0$
$u^2 =$ $8.0^2 = 64$
$v^2 - u^2 =$ $0 - 64 = -64$
$2 \times s =$ $2 \times 16 = 32$
$a =$ $-64 \div 32 = \mathbf{-2.0\ m/s^2}$

Check: $v^2 - u^2 =$ *−64* $\qquad 2\,a\,s =$ *−64*

C. Find the final velocity, v

Worked example C — find the final velocity

A scooter moving at 9.0 m/s accelerates at 2.0 m/s^2 over a distance of 10 m. Calculate its final velocity.

1. Givens

$s = 10$ m
$u = 9.0$ m/s
$v =$ **?**
$a = 2.0$ m/s^2
~~t~~ $=$

2. Equation $v^2 - u^2 = 2\ a\ s$

3. Substitute $v^2 - 9.0^2 = 2 \times 2.0 \times 10$

4. Rearrange $v = \sqrt{9.0^2 + 2 \times 2.0 \times 10}$ *form C*

5. Calculate

$$u^2 = 9.0^2 = 81$$
$$2\,a\,s = 2 \times 2.0 \times 10 = 40$$
$$v^2 = 81 + 40 = 121$$
$$v = \sqrt{121} = \mathbf{11\ m/s}$$

Check: $v^2 - u^2 = 121 - 81 = 40$ and $2\,a\,s = 40$. Same number.

Now try. Add first, and press $\sqrt{\ }$ last.

C1. A cyclist moving at 3.0 m/s accelerates at 3.0 m/s^2 over 12 m. Calculate the final velocity.

Givens

$s = 12$ m
$u = 3.0$ m/s
$v =$?
$a = 3.0$ m/s^2
~~t~~ $=$

Form: $v = \sqrt{u^2 + 2\,a\,s}$

$u^2 = 3.0^2 = 9.0$

$2\,a\,s = 2 \times 3.0 \times 12 = 72$

$v^2 = 9.0 + 72 = 81$

$v = \sqrt{81} =$ ***9.0 m/s***

Check: $v^2 - u^2 =$ *72* $\qquad 2\,a\,s =$ *72*

D. Find the initial velocity, u

Worked example D — find the initial velocity

A motorbike accelerates at $3.0\,\text{m/s}^2$ over $20\,\text{m}$ and reaches $13\,\text{m/s}$. Calculate how fast it was going at the start.

1. Givens

$s = 20\ \text{m}$
$u = \mathbf{?}$
$v = 13\ \text{m/s}$
$a = 3.0\ \text{m/s}^2$
~~t~~ $=$

2. Equation $\quad v^2 - u^2 = 2 \quad a \quad s$

3. Substitute $\quad 13^2 - u^2 = 2 \times 3.0 \times 20$

4. Rearrange $\quad u = \sqrt{13^2 - 2 \times 3.0 \times 20}$ *form D*

5. Calculate

$$v^2 = 13^2 = 169$$
$$2\,a\,s = 2 \times 3.0 \times 20 = 120$$
$$u^2 = 169 - 120 = 49$$
$$u = \sqrt{49} = \mathbf{7.0\ m/s}$$

Check: $v^2 - u^2 = 169 - 49 = 120$ and $2\,a\,s = 120$. Same number.

Now try. Subtract first, and press $\sqrt{\ }$ last.

D1. A car accelerates at $2.0\,\text{m/s}^2$ over $7.0\,\text{m}$ and reaches $8.0\,\text{m/s}$. Calculate its initial velocity.

Givens

$s = 7.0\ \text{m}$
$u = ?$
$v = 8.0\ \text{m/s}$
$a = 2.0\ \text{m/s}^2$
~~t~~ $=$

Form: $u = \sqrt{v^2 - 2\,a\,s}$

$v^2 = 8.0^2 = 64$

$2\,a\,s = 2 \times 2.0 \times 7.0 = 28$

$u^2 = 64 - 28 = 36$

$u = \sqrt{36} = \mathbf{6.0\ m/s}$

Check: $v^2 - u^2 = 28$ $\qquad 2\,a\,s = 28$

E. Calculator round — *one calculator question; answer to 2 significant figures*

Same steps: one calculator line at a time. Round only on the last line, and check with the number before you rounded.

E1. A van moving at $4.0\,\text{m/s}$ accelerates at $1.2\,\text{m/s}^2$ over $50\,\text{m}$. Calculate the final velocity.

Givens

$s = 50$ m
$u = 4.0$ m/s
$v = ?$
$a = 1.2$ m/s^2
~~t~~ =

Form: $v = \sqrt{u^2 + 2\,a\,s}$

$u^2 = 4.0^2 = 16$

$2\,a\,s = 2 \times 1.2 \times 50 = 120$

$v^2 = 16 + 120 = 136$

$v = \sqrt{136} = 11.66 =$ ***12 m/s***

Check: $v^2 - u^2 =$ *120* $\qquad 2\,a\,s =$ *120*